Smart Programming : YouTube Channel

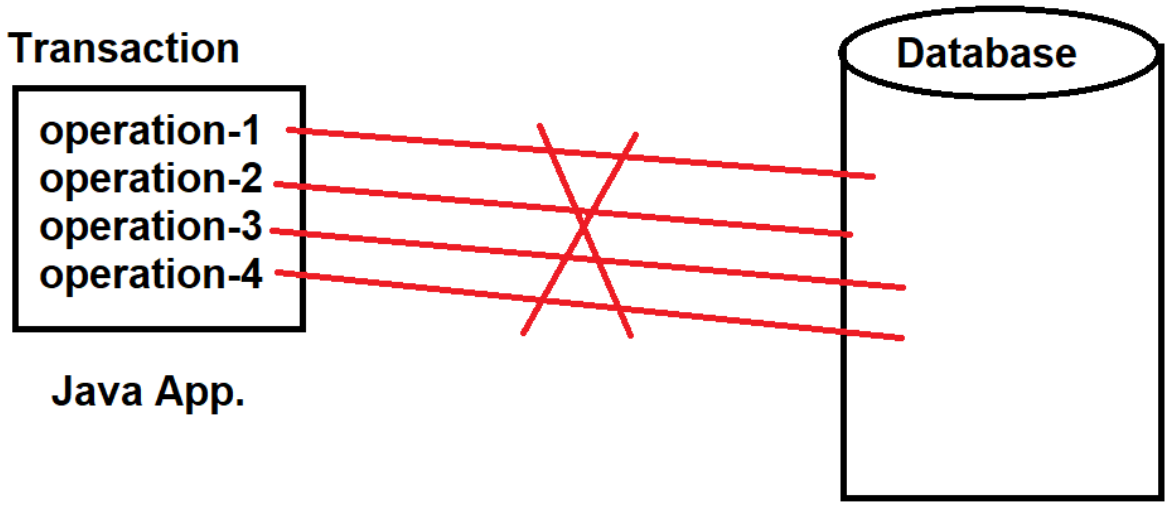
An investment in Knowledge pays the best interest….

JDBC (PART 11)

Transaction Management in JDBC

What is Transaction?

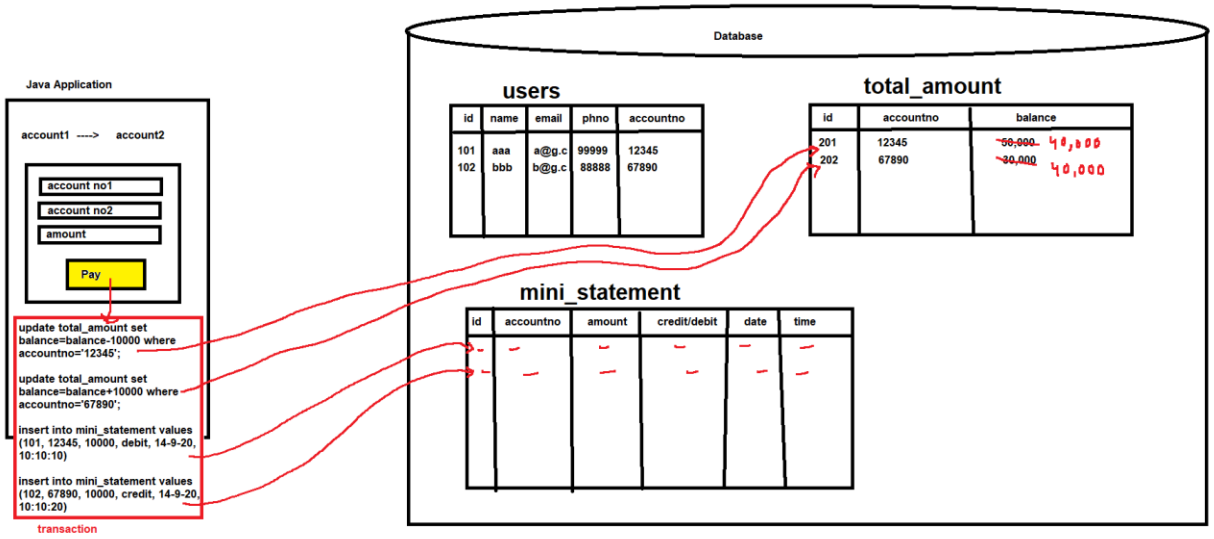
- A group of several operations (SQL queries like insert, update, delete) as a single unit where operations are executed on the principle of either all or none



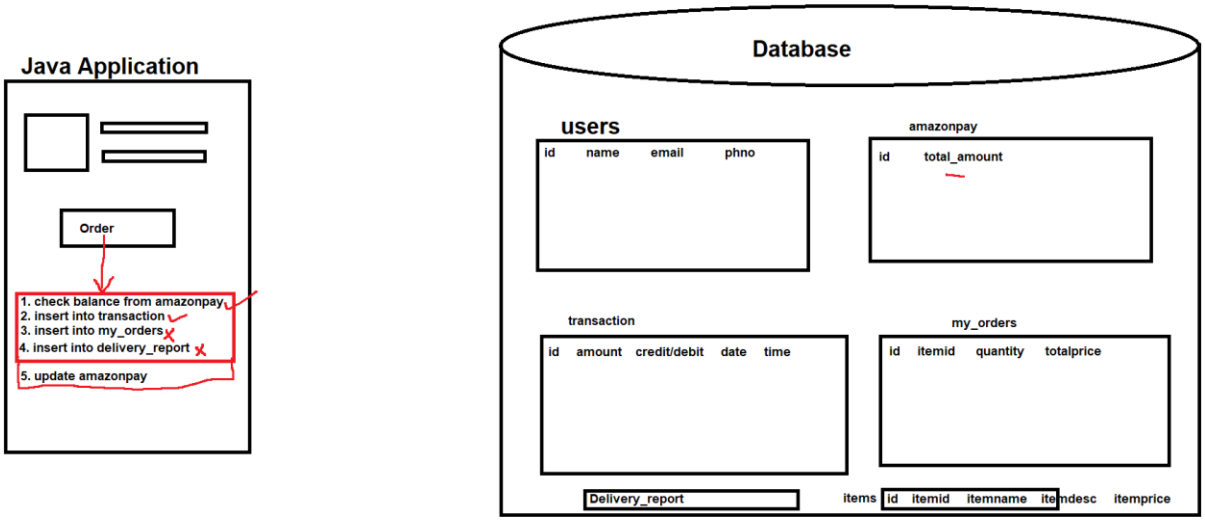
- All operations which are performed from JDBC are by default auto-commit (true)

- Real World App's Example :-

1. Bank Application :-



2. Shopping Cart :- (Amazon, Flipkart)



3. Movie Booking :- 1. transaction

2. seat book details

3. seats info

4. Any booking website :-

How Many types of transactions are there ?

1. Local Transaction : where all the operations are performed on same database

2. Global Transaction : where all the operations are performed on multiple databases

Points to Remember

1. JDBC provides support for local transaction but does not fully support global transaction

2. For global transactions we should go for EJB, JTA, Spring framework etc

Properties of Transaction:-

=> ACID

1. A : Atomicity - Either perform all operations or none
2. C : Consistency - The Database state must be stable
3. I : Isolation - All the transactions must be executed independently, one transaction must not give effect to another transaction
4. D : Durability - The changes of successful transaction occurs even if the system/database failure occurs

How to implement transaction in JDBC ?

To implement transaction, Connection interface provides 3 methods:-

1. setAutoCommit(boolean b)
2. commit()
3. rollback()

Steps to implement transaction:-

1. Set auto-commit false that means we have to insert the values temporary in database, not permanent
2. If all the operations are executed successfully, then commit()
3. If any one operation fails, then execute rollback()

Programming Logic :-

```
con.setAutoCommit(false);
try
{
   operation 1
   operation 2
   operation 3
   operation 4

   if(condition -> true)
   {
         con.commit();
   }
   else
   {
```

```
        con.rollback();
    }
}
catch(Exception e)
{
        con.rollback();
}
```

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